Akoizo Original Practice - 2027 Rotation Form 2

Division C | Dynamic Planet | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Snow depth equals liquid-water depth automatically?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only in spring

B. No

C. Always

D. Only in mountains

Answer: _____

2. (2 points) An oxbow commonly forms through…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only groundwater pumping

B. Meander cutoff

C. Only glacier freezing

D. Only coral growth

Answer: _____

3. (2 points) Every lake mixes twice per year?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if freshwater

B. No

C. Always

D. Only if deep

Answer: _____

4. (2 points) A one-event lab predicts climate?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only for wet regions

B. No

C. Always

D. Only with more sliders

Answer: _____

5. (2 points) Terrace elevation alone establishes exact age?

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only below sea level

B. No

C. Always

D. Only for sandstone

Answer: _____

6. (2 points) A bog is typically associated with…

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only industrial pipes

B. Precipitation-dominated water supply

C. Only seawater tides

D. Only bedrock magma

Answer: _____

7. (2 points) Equal annual rain implies equal reliable supply?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

A. Only for groundwater

B. No

C. Always

D. Only near oceans

Answer: _____

8. (2 points) Migration evidence is strongest from…

Unit: Fluvial landforms and river history.
Worked-case data: Two events share peak discharge but one lasts longer. Why might their geomorphic effects differ?

A. Only river name

B. Comparable positions over time

C. One unlabeled image

D. One water sample

Answer: _____

9. (2 points) Wetlands remove all contaminants forever?

Unit: Lakes, wetlands, and freshwater storage.
Worked-case data: A storage feature lowers peak outflow but releases the same event volume later. Was water destroyed?

A. Only if large

B. No

C. Always

D. Only in summer

Answer: _____

10. (4 points) Why can runoff lag snowfall?

Unit: Freshwater distribution, precipitation, and climate.
Worked-case data: Two storms deliver equal rainfall but one lasts twice as long in the lab. Compare volume and peak.

Answer: _____
_____
_____

11. (2 points) A freshwater catchment receives 120 mm precipitation, loses 40 mm to evapotranspiration and exports 30 mm runoff. Find storage change.

Answer: _____
_____
_____

12. (2 points) A rectangular stream is 6 m wide and 2 m deep with mean velocity 0.5 m/s. Calculate discharge.

Answer: _____
_____
_____

13. (2 points) A river channel is 14 km long between points 10 km apart along the valley. Find sinuosity.

Answer: _____
_____
_____

14. (2 points) A water table drops 2 m over 100 m horizontal distance. Find hydraulic gradient.

Answer: _____
_____
_____

15. (2 points) Use Q=K A i with K=0.001 m/s, A=22 m^2 and i=0.02. Find saturated Darcy discharge.

Answer: _____
_____
_____

16. (2 points) A lake stores 2000 m^3 and exports 200 m^3/day in a steady well-mixed teaching model. Estimate residence time as storage/outflow.

Answer: _____
_____
_____

17. (2 points) An urbanized watershed has more impervious area. Predict one hydrograph change under a comparable storm and explain.

Answer: _____
_____
_____

18. (2 points) A stream bends through its floodplain. Locate likely erosion and deposition relative to the bend.

Answer: _____
_____
_____

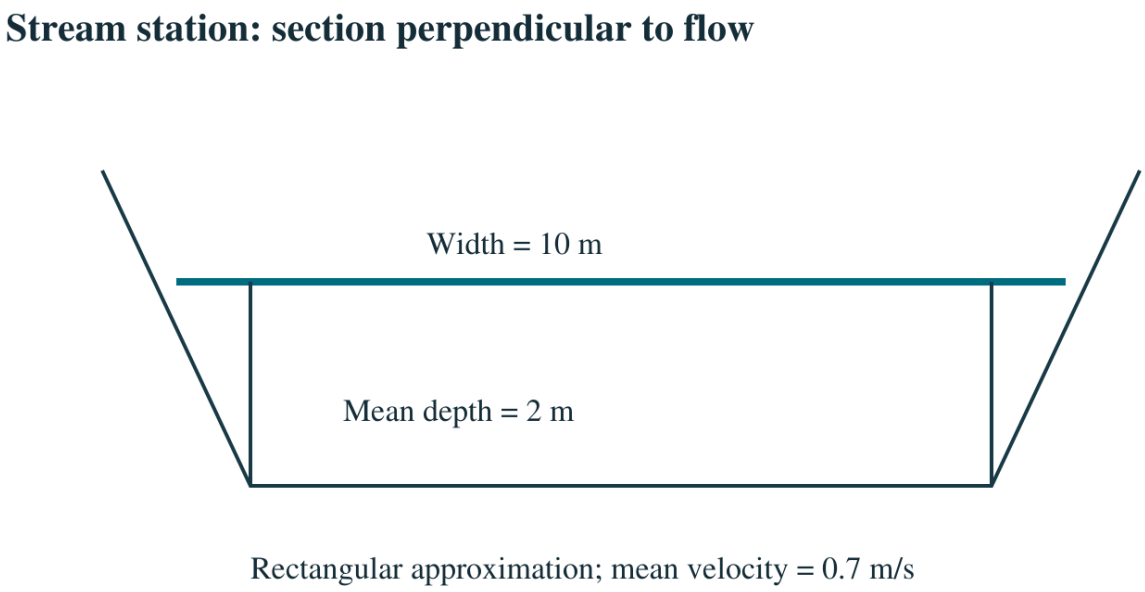
19. (2 points) Two first-order channels join, and no higher-order channel enters. What Strahler order results, and why?

Answer: _____
_____
_____

20. (2 points) Groundwater pumping increases near a connected river. Explain a possible effect on river baseflow.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Estimate wetted cross-sectional area using the indicated rectangular approximation.



F1. Stream cross section and measured velocity (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate stream discharge from this section.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

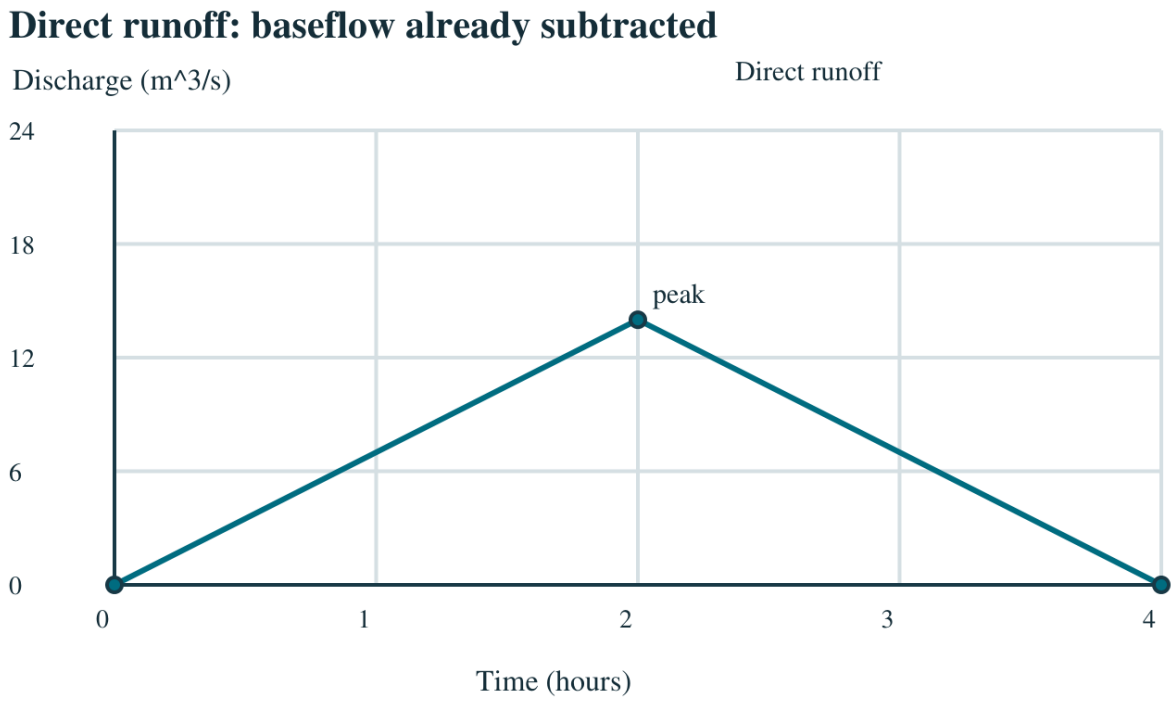
Answer: _____
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23. (2 points) Use Figure F1. The water rises while width and mean velocity remain fixed. Explain the predicted discharge change.

Refer to F1. Stream cross section and measured velocity (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Read the time of peak direct runoff.



F2. Storm hydrograph (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Find the event runoff volume. Convert hours to seconds.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

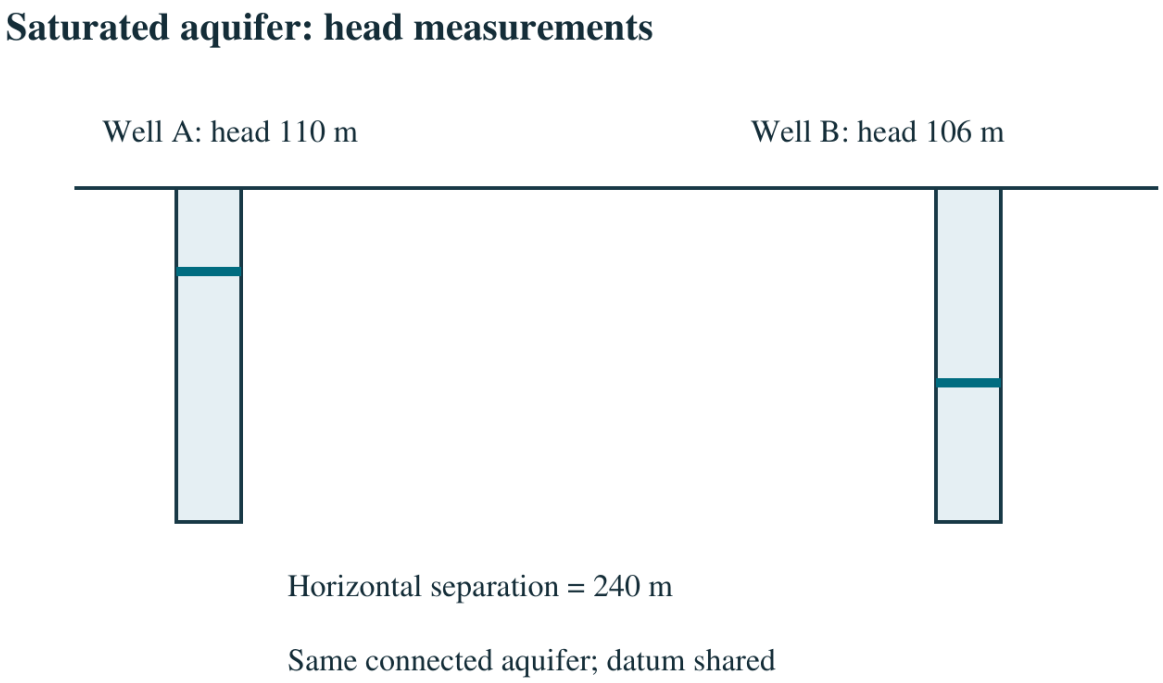
Answer: _____
_____
_____

26. (2 points) Use Figure F2. A wetland spreads the same runoff volume over more time. Describe how its hydrograph could differ.

Refer to F2. Storm hydrograph (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Calculate hydraulic-gradient magnitude.



F3. Groundwater heads in two wells (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Predict the horizontal flow direction and identify the controlling measurement.

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

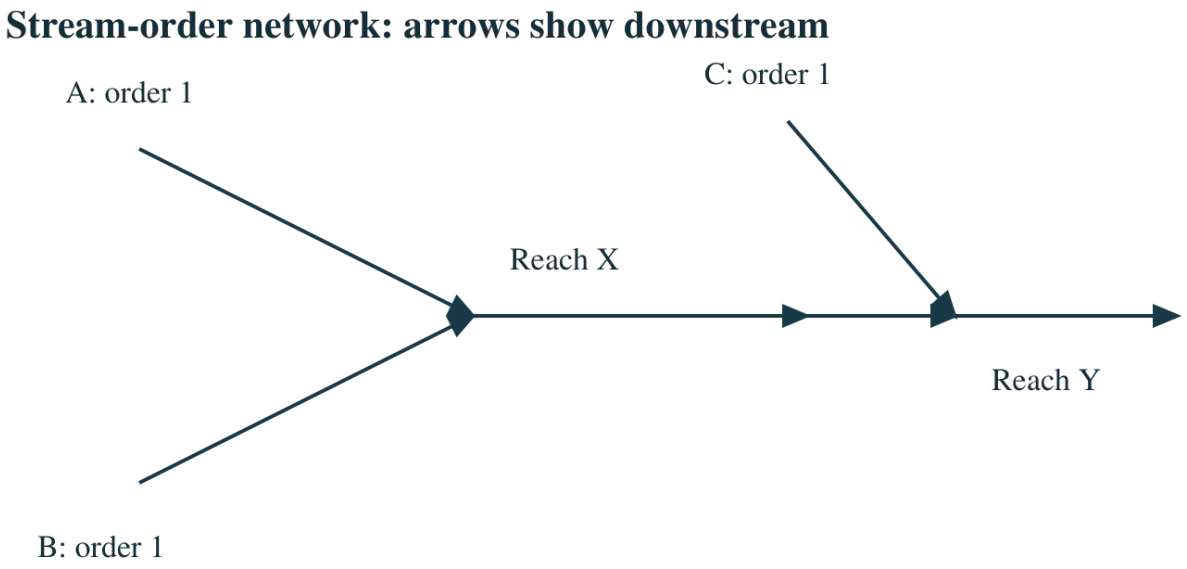
Answer: _____
_____
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29. (2 points) Use Figure F3. Why cannot the diagram alone determine groundwater discharge?

Refer to F3. Groundwater heads in two wells (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Give the Strahler order of reach X.



F4. Tributary network (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Give the Strahler order of reach Y.

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. If C were second order instead, how would Y change?

Refer to F4. Tributary network (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____